

The Impact of Stadium Attendance on Premier League Home Advantage During and Post-COVID-19

Cody Chhea, Reyansh Girdhar, and Rihaan Modi

<https://github.com/UVA-STAT3080/project-sp26-cody-c.git>

Introduction

Home advantage is one of the most common and widely discussed patterns in professional soccer. Teams often perform better when playing at home than when playing away, winning a larger share of their matches in their own stadiums. Several explanations have been proposed for this pattern, including familiarity with the playing environment, reduced travel demands, and support from home fans. In particular, large crowds may benefit home teams by increasing player confidence, creating a more difficult atmosphere for opponents, and influencing referee decisions. The Premier League offers a useful setting in which to examine the importance of fan support. During COVID-19, matches in the Premier League were played with limited or no spectators. After the 2019–20 season was suspended in March 2020, the league resumed in June behind closed doors. Most matches during the 2020–21 season were also played without fans, although limited attendance returned later in the season. These unusual conditions created a natural experiment in which crowd support, one of the main sources of home advantage, was largely removed. This project examines whether home advantage in the Premier League changed during the COVID-19 period and whether it recovered after fans returned. Home advantage is measured using each team's home win percentage for a season. $\text{Home Win \%} = \text{Home Wins} / \text{Home Games Played} \times 100$. The analysis compares home win percentage across two groups of seasons: COVID seasons from 2019-20 through 2020-21 and post-COVID seasons from 2021–22 through 2023–24. The specific research question is: Has the magnitude of home-field advantage in the Premier League significantly changed between the COVID-19 affected seasons and post-COVID-19 eras?

Data Summary

We collected data from fbref, a comprehensive football statistics website, to analyze the impact of the COVID-19 pandemic on home advantage in the Premier League. We merged data of every team from 2016-2024, making our data a population of all the teams in the Premier League, which includes the number of wins at home, total games played at home, and the corresponding seasons. We then categorized the seasons into three groups: Pre-COVID (2016-17 to 2018-19), COVID (2019-20 and 2020-21), and Post-COVID (2021-22 to 2023-24), adding a corresponding column to the dataset. We also calculated each team’s home win percentage — total home wins divided by home games played, multiplied by 100 — and included it as an additional variable.

If we were to compare the home-win percentage from before and after COVID, there can be other factors that could be the cause of any visual changes; not just the attendance factor. As years go by, teams can get better or worse, the quality of the league can change, or even there could be minor league changes that can all affect performance of teams. It is difficult to isolate the absence of fans as the sole cause of performance changes.

However, to answer the question of whether the absence of fans during the COVID-19 seasons and after COVID-19 affected home advantage in the Premier League, our data which encompasses every team-season from 2016 to 2024 allows for a comparison of performance during and after COVID-19. We have clear categorization of matches into distinct chronological phases: a robust “COVID” baseline with no/limited attendance and a “Post-COVID” recovery era. In addition, there is detailed data for every game played at home for each team, every single season, allowing us to transform the raw win/loss records into a standardized “Home Win Percentage” metric that allows for numerical and visual comparisons across the chronological phases.

Exploratory Analysis

Numerical Summary 1: Mean and Standard Deviation of Home Win % by Group

Table 1: Mean and Standard Deviation of Home Win Percentage by COVID Group

Group	N (team-seasons)	Mean Home Win %	SD Home Win %
Pre-COVID	60	47.19	18.18

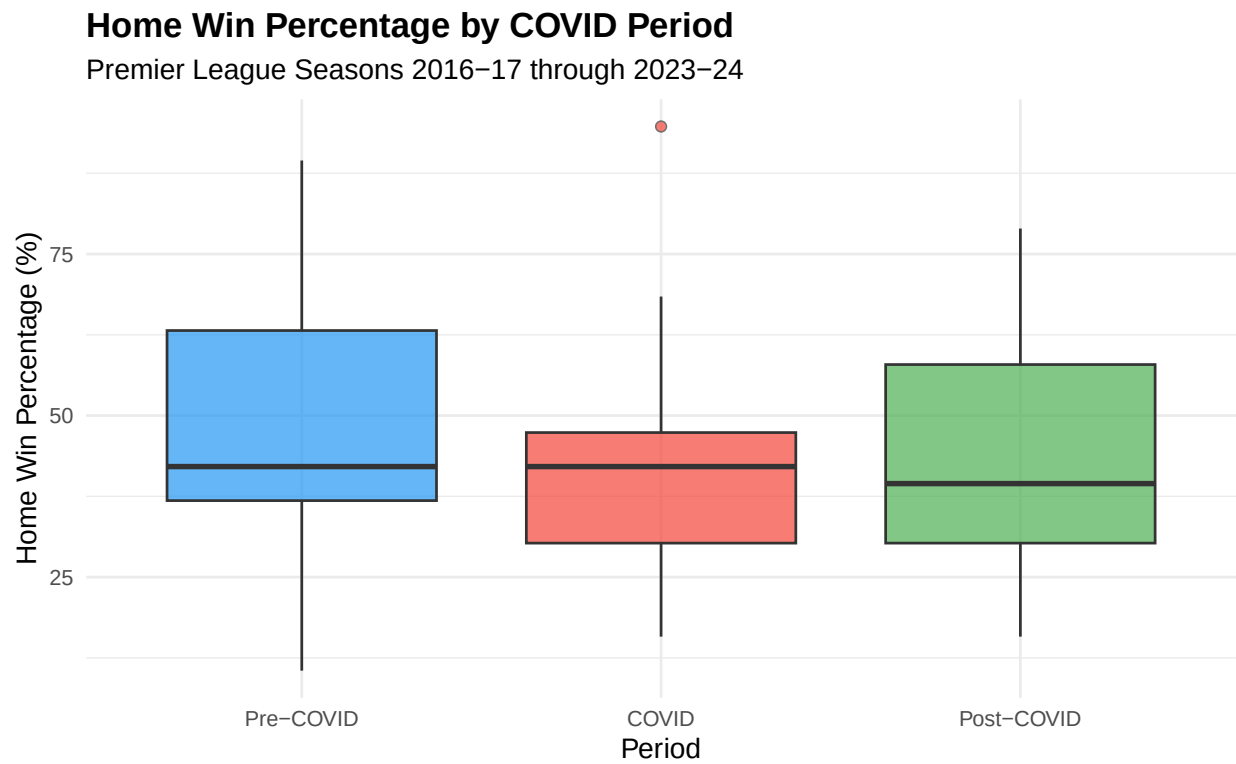
Group	N (team-seasons)	Mean Home Win %	SD Home Win %
COVID	40	41.32	15.66
Post-COVID	60	43.68	18.03

Numerical Summary 2: Five-Number Summary of Home Win % by Season

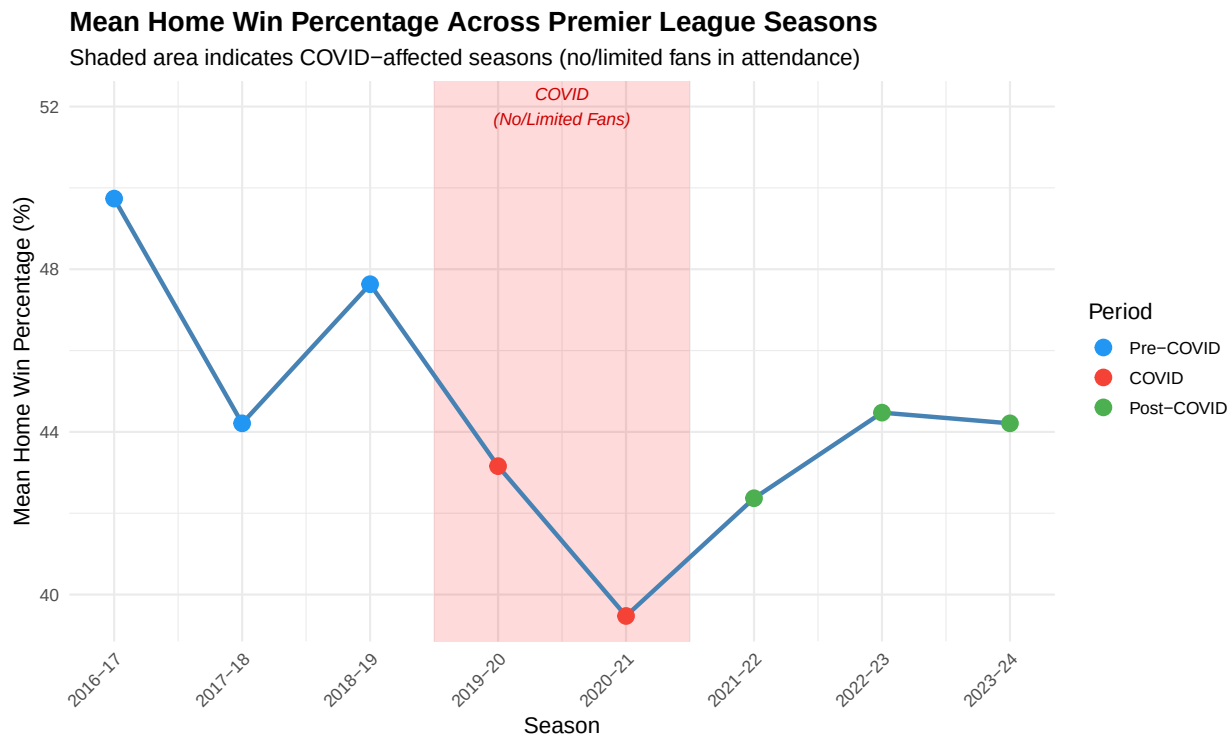
Table 2: Five-Number Summary of Home Win Percentage by Season

Season	Min	Q1	Median	Q3	Max
2016-17	21.1	42.1	42.1	63.2	84.2
2017-18	15.8	31.6	36.8	59.2	84.2
2018-19	10.5	36.8	44.7	56.6	89.5
2019-20	15.8	30.3	42.1	48.7	94.7
2020-21	15.8	30.3	42.1	47.4	68.4
2021-22	15.8	26.3	36.8	53.9	78.9
2022-23	21.1	30.3	39.5	59.2	73.7
2023-24	15.8	35.5	42.1	57.9	73.7

Graphical Summary 1: Boxplot of Home Win % by COVID Group



Graphical Summary 2: Mean Home Win % Trend Over Seasons



Conclusion

The numerical and graphical summaries suggest that home advantage in the Premier League declined during the COVID-affected seasons. Mean home win percentage decreased from 47.2% in the pre-COVID period to 41.3% during the COVID period. The boxplot also shows that the overall distribution of home win percentages shifted downward and became more compressed during the seasons with limited or no fans. The line graph indicates that the largest decline occurred during the 2020–21 season, which was played almost entirely without spectators. In addition, the five-number summaries show that 2020–21 had the lowest maximum home win percentage and one of the narrowest interquartile ranges of any season in the study. These patterns suggest that even the strongest home teams benefited less from playing at home when fans were absent. After fans returned, home win percentage increased to 43.7%, indicating that home advantage partially recovered in the post-COVID seasons. However, the post-COVID mean and distribution remained somewhat below pre-COVID levels. This suggests that fan support may be an important contributor to home advantage, but that other factors may also affect home performance. Changes in team quality, promotions and relegations, tactical adjustments, and lingering effects of the pandemic

could all influence the results. Overall, the exploratory analysis provides evidence that the absence of fans was associated with a reduction in home advantage in the Premier League and that home advantage only partially recovered once fans returned. Because this analysis is descriptive rather than inferential, it cannot determine whether the observed differences are statistically significant. However, the consistent downward pattern across both the numerical and graphical summaries strongly suggests that crowd support plays an important role in home-field advantage.

References

Claude Opus 4.6

Claude Sonnet 4.6

Gemini 3.1 Pro

ChatGPT Plus 5.3

<https://fbref.com/en/comps/9/history/Premier-League-Seasons/> <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0289899>